St. Vincent Pallotti College of Engineering and Technology, Nagpur

Department of Computer Science and Engineering (Data Science)

ACTIVITY REPORT 2025-2026

Title of Project: Railway Reservation System

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1)Introduction

The Railway Reservation System is an essential component of modern transportation management, designed to handle large volumes of passenger bookings efficiently. With the increasing demand for rail travel, manual reservation methods are no longer practical, making automated systems necessary for accuracy, speed, and reliability.

This project presents a simplified Railway Reservation System implemented using the Queue data structure. The system provides core functionalities such as ticket booking, cancellation, and waiting list management, simulating real-world railway reservation operations.

In practical railway systems, managing seat availability and handling waiting lists fairly is crucial. This project applies the FIFO (First In First Out) principle of queues to ensure that passengers in the waiting list are served in the exact order of their arrival, maintaining fairness and transparency.

The main objective of this project is to demonstrate how fundamental data structures like queues can be effectively applied to solve real-world problems involving sequential processing, resource allocation, and dynamic data management.

2)Objective

The objective of this project is to develop a Railway Reservation System using the Queue data structure. The system is designed to simulate real-world railway ticket booking operations, including ticket reservation, cancellation, and waiting list management. It aims to ensure efficient handling of passenger data and fair allocation of seats using appropriate data structure logic.

3)Methodology / Algorithm

1. **System Initialization:**
   - Initialize an array to store confirmed passengers (Booked Tickets).
   - Initialize a queue to manage passengers in the waiting list.
   - Define the maximum number of available seats.
   - Set front and rear of the waiting queue to -1.
2. **User Interaction:**
   - Display a menu-driven interface with options such as Book Ticket, Cancel Ticket, View Status, and Exit.
   - Accept user input and perform operations based on the selected choice.
3. **Ticket Booking Process:**
   - Accept passenger details such as name and age.
   - Generate a unique ticket ID for each booking.
   - If seats are available, add the passenger to the confirmed list (array).
   - If seats are full, add the passenger to the waiting list using the queue (enqueue operation).
4. **Waiting List Management:**
   - Maintain FIFO (First In First Out) order using the queue.
   - Passengers are added to the rear and removed from the front when seats become available.
5. **Ticket Cancellation Process:**
   - Accept the ticket ID to cancel.
   - Search for the passenger in the confirmed list.
   - Remove the passenger by shifting elements in the array.
   - If the waiting list is not empty, remove the first passenger from the queue (dequeue operation) and add them to the confirmed list.
6. **Display Functionality:**
   - Display all confirmed tickets with passenger details.
   - Display the waiting list in proper order.
   - Update and show the system status after every operation.

4) Workflow Diagram

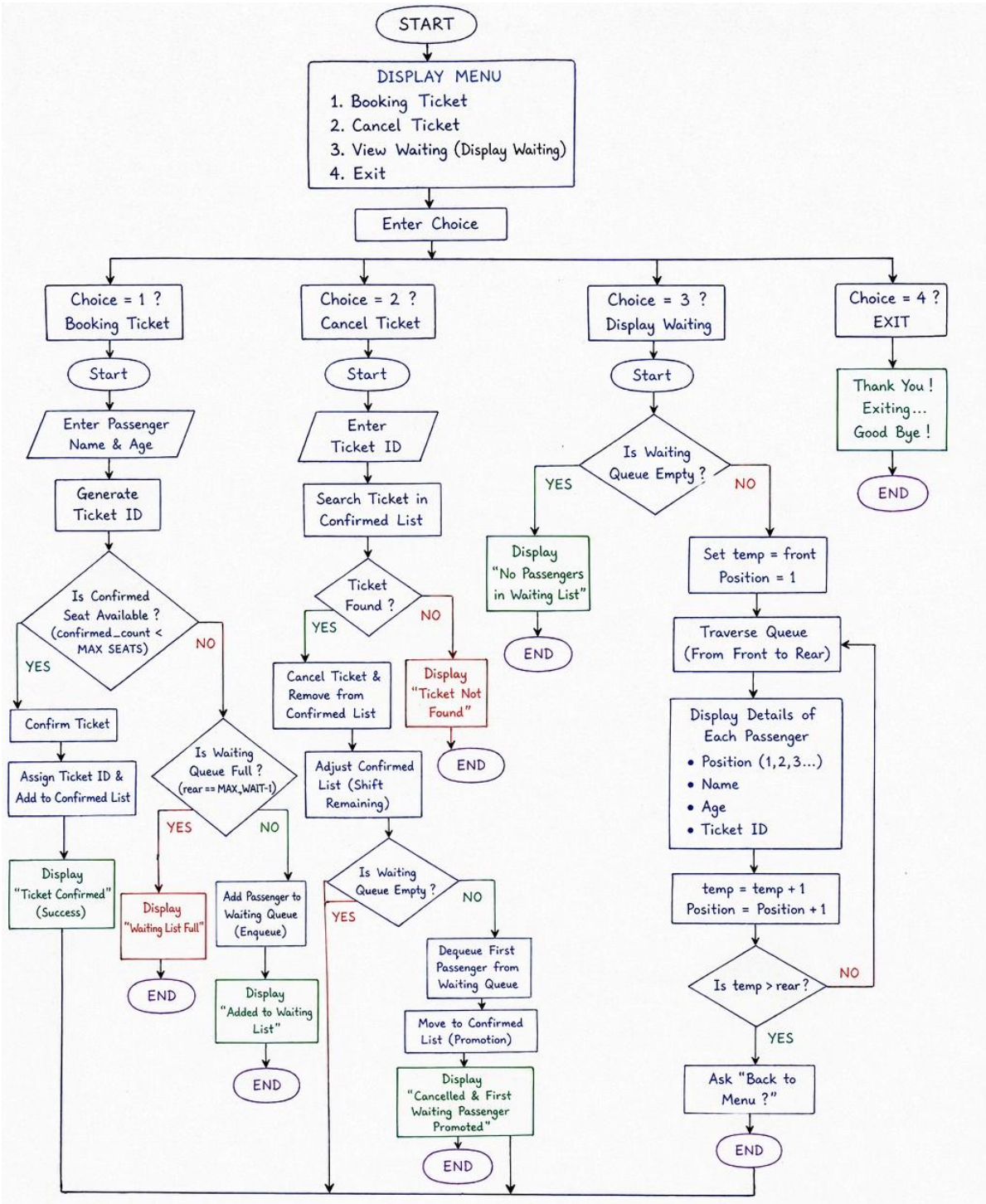

Figure: Workflow of Railway Reservation System using Queue

The above workflow diagram illustrates the step-by-step execution of the Railway Reservation System. It shows how user inputs are processed, how ticket booking and cancellation operations are performed, and how the waiting list is managed using the Queue data structure following the FIFO (First In First Out) principle.

5) Tools & Technologies Used

- **Programming Language:** C
- **Data Structures Used:** Queue (FIFO) for waiting list management, Arrays for storing confirmed bookings
- **Concepts Applied:** Conditional Statements, Loops, Functions
- **Development Environment:** Visual Studio Code / CodeBlocks
- **Platform:** Windows

6)Main Concept of the Project

This project is primarily based on the implementation of the Queue data structure along with arrays for managing passenger records.

- **Queue(FIFO–First_In_First_Out):**
  The queue is used to manage the waiting list of passengers. When all seats are filled, additional passengers are added to the queue. When a ticket is cancelled, the first passenger in the queue is moved to the confirmed list. This ensures fairness, as passengers are served in the exact order of their arrival.

- **Array(Confirmed-Bookings):**
  An array is used to store confirmed passenger details. It allows efficient storage and retrieval of booked tickets, and supports operations like insertion and deletion through shifting elements.

The combination of queue and array helps simulate a simplified but practical railway reservation system, where seat allocation and waiting list handling are managed efficiently and systematically.

7)Output

1. **Main Menu Screen (Starting State)**

```
+--------------------------------+
|   RAILWAY RESERVATION SYSTEM   |
+--------------------------------+
1. Book Ticket
2. Cancel Ticket
3. View Status
4. Exit
Enter your choice:
```

2. **Successful Ticket Booking (Confirmed Case)**

```
+--------------------------------+
|   RAILWAY RESERVATION SYSTEM   |
+--------------------------------+
1. Book Ticket
2. Cancel Ticket
3. View Status
4. Exit
Enter your choice: 1

--- Book Ticket ---
Enter Name: Vishal Masih
Enter Age: 56

[+] SUCCESS: Ticket Confirmed for Vishal Masih! Ticket ID: 1001
```

3. Waiting List Scenario

```
+--------------------------------+
|   RAILWAY RESERVATION SYSTEM   |
+--------------------------------+
1. Book Ticket
2. Cancel Ticket
3. View Status
4. Exit
Enter your choice: 1

--- Book Ticket ---
Enter Name: Lamine Yamal
Enter Age: 18

[*] Train is full. Added Lamine Yamal to the Waiting List.

+--------------------------------+
|   RAILWAY RESERVATION SYSTEM   |
+--------------------------------+
1. Book Ticket
2. Cancel Ticket
3. View Status
4. Exit
Enter your choice: 3

==================================
      CURRENT TRAIN STATUS
==================================
CONFIRMED TICKETS (3/3):
  [1001] Vishal Masih (Age: 56)
  [1002] Lionel Messi (Age: 38)
  [1003] Cristiano Ronaldo (Age: 42)

WAITING LIST:
  [1004] Lamine Yamal (Age: 18)
==================================
```

4. View Status Output

```
+-----------------------------------+
|   RAILWAY RESERVATION SYSTEM    |
+-----------------------------------+
1. Book Ticket
2. Cancel Ticket
3. View Status
4. Exit
Enter your choice: 3

===================================
      CURRENT TRAIN STATUS
===================================
CONFIRMED TICKETS (3/3):
  [1001] Vishal Masih (Age: 56)
  [1002] Lionel Messi (Age: 38)
  [1003] Cristiano Ronaldo (Age: 42)

WAITING LIST:
  [1004] Lamine Yamal (Age: 18)
  [1005] Kylian Mbappe (Age: 27)
  [1006] Earling Haaland (Age: 24)
===================================
```

5. Ticket Cancellation

```
+-------------------------------+
|   RAILWAY RESERVATION SYSTEM   |
+-------------------------------+
1. Book Ticket
2. Cancel Ticket
3. View Status
4. Exit
Enter your choice: 2

--- Cancel Ticket ---
Enter Ticket ID to cancel: 1001

[-] Ticket ID 1001 for Vishal Masih has been cancelled.
[^] ALERT: Lamine Yamal has been promoted from the Waiting List to Confirmed!
```

6. Waiting List Promotion

```
+--------------------------------+
|   RAILWAY RESERVATION SYSTEM   |
+--------------------------------+
1. Book Ticket
2. Cancel Ticket
3. View Status
4. Exit
Enter your choice: 2

--- Cancel Ticket ---
Enter Ticket ID to cancel: 1001

[-] Ticket ID 1001 for Vishal Masih has been cancelled.
[^] ALERT: Lamine Yamal has been promoted from the Waiting List to Confirmed!

+--------------------------------+
|   RAILWAY RESERVATION SYSTEM   |
+--------------------------------+
1. Book Ticket
2. Cancel Ticket
3. View Status
4. Exit
Enter your choice: 3

==================================
      CURRENT TRAIN STATUS
==================================
CONFIRMED TICKETS (3/3):
  [1002] Lionel Messi (Age: 38)
  [1003] Cristiano Ronaldo (Age: 42)
  [1004] Lamine Yamal (Age: 18)

WAITING LIST:
  [1005] Kylian Mbappe (Age: 27)
  [1006] Earling Haaland (Age: 24)
==================================
```

7. Final Updated Status

```
+--------------------------------+
|   RAILWAY RESERVATION SYSTEM   |
+--------------------------------+
1. Book Ticket
2. Cancel Ticket
3. View Status
4. Exit
Enter your choice: 3

==================================
      CURRENT TRAIN STATUS
==================================
CONFIRMED TICKETS (3/3):
  [1002] Lionel Messi (Age: 38)
  [1003] Cristiano Ronaldo (Age: 42)
  [1004] Lamine Yamal (Age: 18)

WAITING LIST:
  [1005] Kylian Mbappe (Age: 27)
  [1006] Earling Haaland (Age: 24)
==================================
```

8) Result

The Railway Reservation System was successfully designed and implemented using the Queue data structure in the C programming language. The system efficiently performs core operations such as ticket booking, cancellation, and waiting list management.

During execution, the program correctly allocates seats to passengers until the maximum capacity is reached. Once all seats are filled, additional passengers are automatically added to the waiting list, maintaining a First In First Out (FIFO) order using the queue. This ensures fairness and proper sequencing of requests.

The cancellation functionality operates effectively by removing the selected passenger from the confirmed list and immediately promoting the first passenger from the waiting list to confirmed status. This demonstrates proper handling of dynamic data and real-time updates within the system.

Additionally, the system provides accurate and updated status displays for both confirmed tickets and waiting list passengers after each operation, ensuring clarity and usability.

Overall, the project successfully demonstrates the practical application of the Queue data structure in solving real-world problems related to reservation systems, resource allocation, and sequential data processing.

Faculty Coordinator:

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